

Euler's Totient Function

$\varphi(n)$

Definition: Euler's Totient Function, $\varphi: \mathbb{N} \rightarrow \mathbb{N}$

with

$$\varphi(n) = \text{card}(\{m \in \mathbb{N} \mid 1 \leq m < n \mid \gcd(m, n) = 1\})$$

examples: $\varphi(5) = |\{1, 2, 3, 4\}| = 4$

$\varphi(8) = 4$, $\varphi(15) = 8$

proposition: For a prime p we have $\varphi(p) = p - 1$.

proof: Notice the only divisors of p are 1 and p .

Let $1 \leq m < p$ then $p \nmid m$ and $2 \leq m < p$

doesn't divide p , only $m = 1$. $\Rightarrow \gcd(m, p) = 1$

□

proposition: If $\gcd(m, n) = 1$ then $\varphi(mn) = \varphi(m) \cdot \varphi(n)$.

example: $15 = 3 \cdot 5$ $\varphi(15) = 8$, $\varphi(3) = 2$, $\varphi(5) = 4$

① ② 3
④ 5 6
⑦ 8 9
10 ⑪ 12
⑬ ⑭ 15

obs: There are $\varphi(3)^2$ columns with relatively primes to 3. In each of these columns there are $\varphi(5)^4$ rel. prime to 5.

$$\varphi(15) = \varphi(3) \cdot \varphi(5) = 8$$

proof: Consider

1	2	...	m-1	m
m+1	m+2	...	2m-1	2m
...	...	...	...	...
(n-1)m+1	(n-1)m+2	...	nm-1	nm
nm-m+1	nm-m+2	...	nm-1	nm

observe: there are $\varphi(m)$ columns made completely of numbers relatively prime to m .

Say we are in column k : $k, m+k, \dots, nm-m+k$

claim: each of these columns contains exactly $\varphi(n)$ numbers relatively prime to n .

Take 2 numbers from column k , say $lm+k$ and $l'm+k$

note: $lm+k \equiv (l'm+k) \pmod{n}$, because

$$\Leftrightarrow (l-l')m \equiv 0 \pmod{n}$$

$$\Rightarrow n \mid l-l' \Rightarrow l=l', \text{ since } l-l' < n$$

$\Rightarrow$ each column contains $\varphi(n)$ numbers relatively prime to n .

$$\Rightarrow \varphi(nm) = \varphi(n) \varphi(m) \quad \square$$

(Short) Version for Primes

proposition: Let p, q primes $\Rightarrow \varphi(p \cdot q) = \varphi(p) \cdot \varphi(q)$

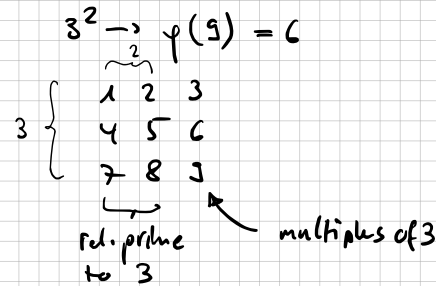
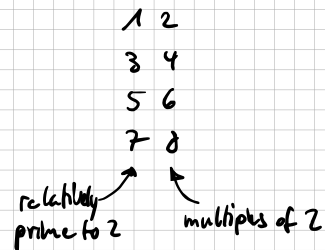
note: $\begin{matrix} 1 \cdot p & 1 \cdot q \\ 2 \cdot p & 2 \cdot q \\ \vdots & \vdots \\ (q-1)p & (q-1)q \\ q \cdot p & p \cdot q \end{matrix} \rightarrow$ cause p, q primes there are $q-1$ numbers divisible by p and $p-1$ different numbers divisible by q .

Further $p \cdot q$ is divisible by p and q . So there are $p \cdot q$ numbers of which $pq - (q-1 + p-1 + 1)$ are relatively prime to p or q :

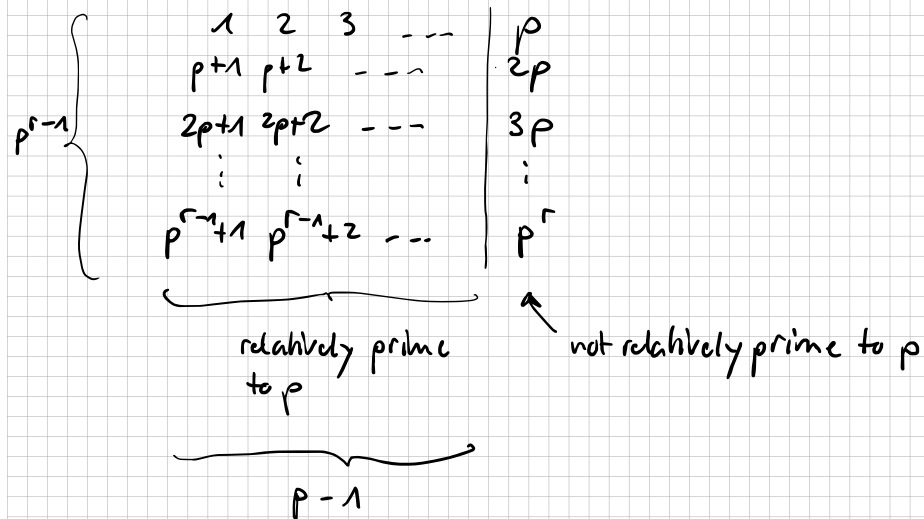
$$pq - (q + p - 1) = pq - q - p + 1 = (p-1)(q-1) = \varphi(p) \cdot \varphi(q) \quad \square$$

proposition: For a prime p , we have $\varphi(p^r) = p^r - p^{r-1}$

examples: $2^3 \rightarrow \varphi(8) = 4$



proof: Arrange $1, 2, 3, \dots, p^{r-1}, p^r$ in an array:



$$\Rightarrow \varphi(p^r) = p^{r-1} \cdot (p-1) = p^r - p^{r-1} \quad \square$$

Theorem: If $n = p_1^{r_1} \dots p_k^{r_k}$ then

$$\varphi(n) = n \left(1 - \frac{1}{p_1}\right) \dots \left(1 - \frac{1}{p_k}\right)$$

$$\begin{aligned} \text{proof: } \varphi(n) &= \varphi(p_1^{r_1} \dots p_k^{r_k}) = \varphi(p_1^{r_1}) \cdot \dots \cdot \varphi(p_k^{r_k}) \\ &= (p_1^{r_1} - p_1^{r_1-1}) \cdot \dots \cdot (p_k^{r_k} - p_k^{r_k-1}) \\ &= p_1^{r_1} \left(1 - \frac{1}{p_1}\right) p_2^{r_2} \left(1 - \frac{1}{p_2}\right) \dots p_k^{r_k} \left(1 - \frac{1}{p_k}\right) \\ &= p_1^{r_1} \dots p_k^{r_k} \left(1 - \frac{1}{p_1}\right) \dots \left(1 - \frac{1}{p_k}\right) \\ &= n \cdot \left(1 - \frac{1}{p_1}\right) \dots \left(1 - \frac{1}{p_k}\right) \quad \square \end{aligned}$$

Bem: Man muss nur die Primfaktoren von n kennen, aber nicht die Anzahl mit der sie vorkommen.